BM innovations GmbH
incremental changes in thinking

Pam Sensor Device

BLE Interface Specification

Date: 27th of Feb 2024
Revision: 1.8

The Pam sensor device contains a Bluetooth low energy radio for wireless communication, allowing it to be part of a short distance body area network e.g. in combination with a smartphone. This document specifies the communication part.

Pam Sensor Device Functionality

The device contains a sensor for 3 axis acceleration measurement. Measured and also pre-processed data can be recorded in several ways in the device, but also transmitted wirelessly via Bluetooth low energy (BLE) to devices supporting this protocol.

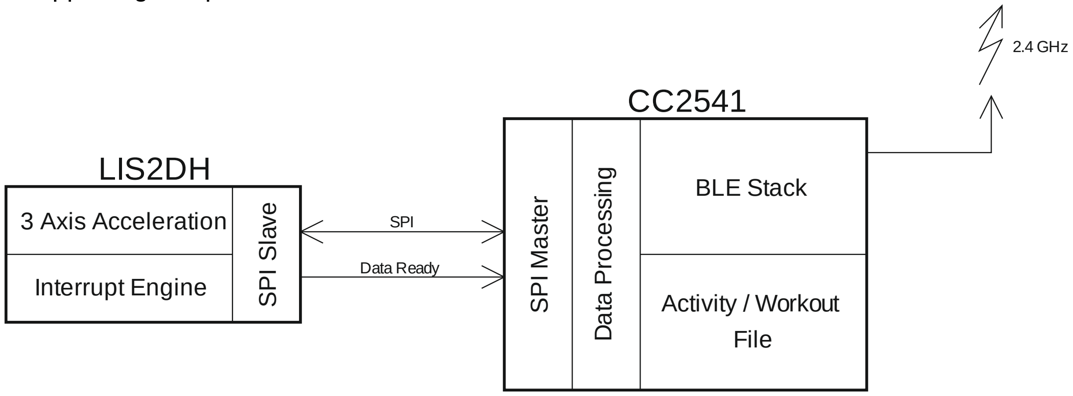

Figure 1: Block Diagram

Device Operating Modes

The sensor device can be set to a standby/transportation mode where only the acceleration sensor is enabled causing the device to leave this mode as soon as an acceleration passes the standby threshold and an activity mode where steps, activity score and activity zone summary will be determined.

Files for Data Recording

Two independent files are available on for downloading from Pam. The first one contains activity summary for current day, the other one contains detailed activity information with 15-minutes timestamps.

Activity Measurement

Whenever the sensor is not in standby mode and detects motion the activity score, activity zone summary and steps will be determined for the current day and if enabled sent via notifications to a bonded peer device. Steps and activity score will also be recorded in the activity file every minute.

BLE Communication Overview

The Pam sensor device supports the standard Battery Service (UUID 0x180F) and Device Information Service (UUID 0x180A).

For being able to configure the sensor and get further data determined on the sensor side, 2 customized sensor services are implemented:

Service / Characteristic	UUID	Bytes	Properties	Notes
Pam Device Service	2000			
Time Date	2001	4	Read/Write	uint32 value. Time in UTC format in seconds (since 00:00:00: 01.01.1970)
Setup (Test Data)	2002	8	Read/Write	Byte 0: Reserved Byte 1: Threshold Sensor activation threshold in 10mg: 1 (10mg) to 100 (1g), default is 180mg Byte 2: Threshold Sensor deactivation threshold in 10mg: 1 (10mg) to 100 (1g), default is 70mg Byte 3: Time Sensor deactivation time in 10s: 1 (10s) to 200 (2000s), default is 120s Byte 4: Interval Advertising interval, see "Understanding Advertising Intervals" below, default is 546.25ms Byte 5: Reserved Byte 6: Reserved Byte 7: Interval Connection interval in 12.5ms: 1(12.5ms) to 237 (2962.5ms), default is 62ms
Standby (Test for Production)	2003	1	Read/Write	Byte 0: 0 for standby, otherwise starts test modes (tbd) where data will be exchanged using Test Data.

Service / Characteristic	UUID	Bytes	Properties	Notes
Activity Service	2100			
Activity Data	2101	8	Notify	All activity data will be sent to the peer device via notifications. The notification interval is equal to the requested slow connection interval. A description of the Activity Data format can be found below.
Activity File	2102	2	Write	Byte 0: Low byte of file request command Byte 1: High byte of file request command File request format depends on which file is requested – day data file or detailed file. Request format is described below. Requested data will be returned in the Activity Download characteristic. To start the data transfer the Download notification has to be enabled. For an immediate start this can already be done before requesting the download or even directly after bonding.
Activity Download	2103	102	Notify	Block(s) of requested activity file data. First byte contains low byte of block number, second byte contains high byte of block number followed by up to 18 data bytes. A description of the activity file format can be found below.

For any values being represented in more than 1 byte the least significant byte will be stored at the lowest address / sent first and the most significant byte will be stored at the highest address / sent last (little endian).

The UUIDs for the custom services have the following 128 bit values where XXXX has to be replaced with the corresponding UUIDs listed in the table above:

99DBXXXX-AC2D-11E3-A5E2-0800200C9A66

Understanding Advertising Intervals

Lower 4 bits [0:3] of value specify a time according to the following table:

1:	152.5 ms
2:	211.25 ms
3:	318.75 ms
4:	417.5 ms
5:	546.25 ms
6:	760 ms
7:	852.5 ms
8:	1022.5 ms
9:	1285 ms

Higher 4 bits [4:7] of value specify a multiplier for the specified time.

Example: Value = 0x47 -> 4 x 852.5 ms = 3410 ms

Activity Data Format

Activity data sent via notifications to a bonded peer device contain steps, activity score and activity zone summary for the current day (starting at midnight):

- Notification packet has the next structure:

```
typedef struct
{
  uint16 handle;  //!< Handle of the attribute that has been changed (must be   first field)
  uint8 len;                    //!< Length of value
  uint8 value[ATT_MTU_SIZE-3]; //!< New value of the attribute
} attHandleValueNoti_t;
```

- Data are packed in the *value* field:
  Byte 0: Low byte of steps
  Byte 1: High byte of steps
  Byte 2: Low byte of activity score

Byte 3: High byte of activity score
Byte 4: Lower 8 bits of time in zone 3 (sport) in minutes
Byte 5: Upper 1 bits of time in zone 3 (sport) in minutes
Lower 7 bits of time in zone 2 (health) in minutes
Byte 6: Upper 4 bits of time in zone 2 (health) in minutes
Lower 4 bits of time in zone 1 (living) in minutes
Byte 7: Upper 8 bits of time in zone 1 (living) in minutes

File request format.

The MSB of the Byte1 indicates, which file must be downloaded: 0 for Day Data File and 1 for Detailed File.

- When Day Data File is requested, Byte0 indicated the number of days that should be sent.
  Max value = 31 days. If 0 value is written – 1 day will be sent.
- When Detailed Data File is requested, 15 lower bits of [Byte1:Byte0] represent a number of 15-minutes timestamps back in time from the current moment. When device is active, new timestamp is recorded every minute. For example, if data for the last 600 minutes is required, then a value of 40 (600 / 15 = 40) should be written to the lower 15 bits of the Activity File characteristic. If the device was active all the time, Detailed File will contain 40 records; if device was active not continuously during requested period, only existing timestamps will be included in the detailed file. In other words, Detailed File will contain all the existing data for requested period, but number of returned records does not have to be equal to (can be less than) the requested number of timestamps.
  Max value = 3072 (32 days * 24 hours * 15 minutes). Max number of returned records = 10000 (max size of the Detailed File).

To make the above easier to understand, we provide a few EXAMPLES:
- To request a Day Date File of the last 20 days, the command to send is 0x1400.
- To request a Detailed Data File of the last 15 hours, we have to ask for 15x4 quarter hours, so the command to send is 0x3C80. The second byte, 0x80, in binary is 1000 0000: so the highest byte is set to 1, which means the detailed file.

File Header Format

The first file block contains the File Header. In cases, when Pam Detailed Data File is being sent in two parts – first block in both parts contains File Header related to corresponding part. File Header block is 6 bytes long and contains Block Number (which is always 0x0000), File Size and Base Date. File Size indicates the number of bytes that follow this header (it may be number of bytes in the whole file or in the file part, which is sent after this header). The Base Date for Detailed Data File is calculated as:

baseData = EpochTime / (24 * 60 * 60)

and it means the day number after 01.01.1970.

For Day Data File it will contain the current date, since data for up to 31 last days will be sent, including current day.

```
struct
{
      uint16      blockNumber;      // always 0x0000 for file header
      uint16      fileSize;         // Lower 15 bits are size of file to send.
                                    // Set MSB if file will be sent in two parts
      uint16      baseDate;
}
```

Note: MSB fileSize field indicates whether file will be sent in two parts. If it is set, then fileSize represents size of the first part of the file. Second part will be sent with its own file header. For all `uint16` fields endianness – little endian.

Again it is helpful to give an EXAMPLE. When we ask for the last hour of the Detailed Data file, the command is 0x0480. The result is:

```
0x0000b400a54c len=6
0x0100ff5e00001f5f00013f5f00025f5f00047f5f00019f5f0002bf5f010cdf5f0002ff5f041b1f60030f3f6001025f6000
007f6003159f600001bf600000df600000ff6000001f6106283f6100025f61101d7f6100099f610000bf610000df610000ff
610214 len=102
0x02001f6201083f6200005f6200007f6200009f620000bf620000df620000ff6200001f6300001f6500013f6500005f6500
007f6500009f650000bf650000df650003ff6500011f6600003f6600005f660000 len=82
```


Let us parse this data. The first line: first two bytes indicate the message number: 0. The second two bytes are the filesize: 0x00b4 = 180 bytes. The last two bytes are the basedate: 0x4CA5 = 19621. So the epoch is 19621x24x3600=1695254400 corresponding to September 21, 2023.
The second line starts with two bytes for the message counter (1 in this case) and then the string of one minute records of 4 bytes each. How these records have to be interpreted is explained below.

Activity File Format

- **Pam Day Data Packing:**

ST_15	ST_14	ST_13	ST_12	ST_11	ST_10	ST_9	ST_8	ST_7	ST_6	ST_5	ST_4	ST_3	ST_2	ST_1	ST_0
bit 63							bit 56	bit 55							bit 48

PS_12	PS_11	PS_10	PS_9	PS_8	PS_7	PS_6	PS_5	PS_4	PS_3	PS_2	PS_1	PS_0	SZ_8	SZ_7	SZ_6
bit 47							bit 40	bit 39							bit 32

SZ_5	SZ_4	SZ_3	SZ_2	SZ_1	SZ_0	HZ_9	HZ_8	HZ_7	HZ_6	HZ_5	HZ_4	HZ_3	HZ_2	HZ_1	HZ_0
bit 31							bit 24	bit 23							bit 16

LZ_10	LZ_9	LZ_8	LZ_7	LZ_6	LZ_5	LZ_4	LZ_3	LZ_2	LZ_1	LZ_0	DI_4	DI_3	DI_2	DI_1	DI_0
bit 15							bit 8	bit 7							bit 0

ST_[15:0] – Steps
LZ_[10:0] – Living Zone
HZ_[9:0] – Health Zone
SZ_[8:0] – Sport Zone
PS_[12:0] – PAM Score
DI_[4:0] – Date Index

```
typedef struct {
    uint8    date_index   : 5;
    uint16   living_zone  : 11;
    uint16   health_zone  : 10;
    uint16   sport_zone   : 9;
    uint16   pam_score    : 13;
    uint16   steps;
} pamData_t;
```

NOTE:
Activity Day Data File contains steps, activity score and activity zone summary for the last 31 days including current day. File is stored as a rotating buffer so that the older data is overwritten with the newer one.

- **Pam Detailed Data Packing:**

```
typedef struct {
    uint8_t date_index  : 5;
    uint16_t ts_index   : 11;
    uint8_t steps;
    uint8_t pam_score;
} pamDetailedData_t;
```

NOTE:
- The file size is 10000 minutes max: at 12 hours active per day it will cover 13 days.
- File is stored as a rotating buffer: when offset of the next record from file start date exceeds the valid range, older records are rewritten with new data.
- PAM score should be divided by 16 to get float value (0 … 31.875).
- File header will contain its starting date in UTC format so that time and date of each record can be calculated as shown in the example. Max DI = 31.
- EXAMPLE:

Suppose, Starting date (baseDate or BD) is 10.08.2014. The very first record will have DI = 0.
Suppose, some record has Date Index (DI) = 8, Timestamp Index (TI) = 500.
Then, Date = BD + DI = 18.08.2014;
Time = TI = 500 minutes = 8 hours 20 minutes, which is +2 hours in our zone (Amsterdam).
So, this record corresponds to 10:20 am 18.08.2014.

Change Log

Revision	Content	Made by	Date
1.0	Initial Revision.	Glib Dovgych	21.04.2015
1.1	Activity file requesting and downloading format changed.	Glib Dovgych	09.05.2015
1.2	Changed description of the Byte7 in Setups characteristic	Glib Dovgych	13.08.2015
1.3	Changed the format of Detailed File request.	Glib Dovgych	22.09.2015
1.4	Added description of the file header.	Glib Dovgych	10.11.2015
1.5	Added explanation for PAM file header.	Glib Dovgych	23.01.2017
1.6	Changed detailed file to 1 min interval.	Rostyslav Stolyar	24.09.2023
1.7	Some correction for new 1 min interval done.	Rostyslav Stolyar	26.09.2023
1.8	Clean up the setup profile	Rostyslav Stolyar	27.02.2024
	`		